

# Relationally Grounded Latent World Models for Autonomous Driving

Fabian Schmidt<sup>1,2</sup>, MarkusENZweiler<sup>1</sup>, Abhinav Valada<sup>2</sup>

**Abstract**—Latent world models learn predictive representations for autonomous driving, but the relational semantics these states preserve often remain implicit. We investigate whether traffic scene graphs can serve as *privileged semantic supervision* for latent world representations. Building on LAW, we construct actor-centric scene graphs from nuScenes 3D annotations, encode their serialized relational structure using a frozen text embedding model, and align the visual latent representations with this semantic target during training. We remove the supervision branch at inference, so it requires neither scene graphs nor 3D annotations and adds no test-time computation. On nuScenes, our method reduces average trajectory L2 error from 0.661 to 0.622 (5.9%) and collision rate from 0.456 to 0.217 (52.4%) relative to our retrained LAW baseline. It also outperforms an unstructured caption-style semantic target, supporting the benefit of explicit relational structure for latent world-model representation learning.

## I. INTRODUCTION

End-to-end autonomous driving has evolved from unified perception–planning architectures toward foundation model-based agents with increasingly rich semantic representations. Methods such as UniAD [1] and VADv2 [2] jointly learn perception and planning representations, while language-aware approaches such as LMDrive [3], DriveLM [4], SimLingo [5], and OpenDriveVLA [6] increasingly couple semantic reasoning with driving actions. LAD-Drive [7] further bridges the gap between high-level semantic intent and continuous planning through an action-aware diffusion decoder. Together, these works highlight the value of semantically structured representations for driving while generally retaining semantic reasoning in the deployed model.

A complementary line of research investigates world models that learn predictive representations of driving scenes. Generative driving world models predict future visual observations [8]–[11], whereas latent world models reason in compact feature spaces that integrate efficiently with downstream planning. LAW [12] predicts future visual latents conditioned on the current scene representation and ego motion, while WoTE [13] predicts future BEV states for trajectory evaluation. World4Drive [14] introduces spatial-semantic priors, and recent methods increasingly couple latent world modeling with planning, reasoning, and structured latent representations [15]–[22].

Despite this progress, the relational semantics a latent world state should preserve often remain implicit. Predicting future latent states encourages temporal scene modeling, while geometric priors and planning objectives provide spatial and task-level supervision, but do not explicitly constrain relationships

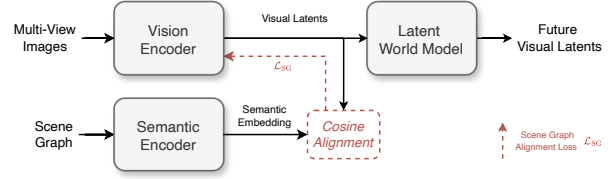


Fig. 1. Overview of the proposed training-only relational grounding. Scene-graph semantics are aligned with visual latents through an additional cosine-alignment loss to encourage interaction-relevant relational structure in the learned representation, while the world-model inference pathway remains unchanged.

between traffic participants. For driving, such relationships are particularly important, since an actor’s relevance depends not only on its appearance but also on its relative position and interaction with the ego vehicle [23]. GraphWorld [24] explicitly incorporates an ego-centric interaction graph into its latent world representation and planning architecture, while GraphPilot [25] shows that structured scene-graph supervision benefits language-based driving. However, these approaches either embed relational structure in the model or target it for language-based reasoning. Whether relational scene structure can instead serve as *privileged semantic supervision* for predictive world representations without requiring scene graphs at inference remains largely unexplored.

In this work, we investigate this question by relationally grounding the latent state of LAW using traffic scene graphs available only during training. We serialize actor-centric graphs constructed from nuScenes [26] 3D annotations into structured relational text and embed them using a frozen text encoder. A lightweight alignment head maps LAW’s visual latents into the same semantic space and aligns them with the scene-graph representation, as shown in Fig. 1. After training, the complete supervision pathway is discarded, requiring neither scene graphs, 3D annotations, nor additional computation at inference. The objective is not specific to LAW and, in principle, applies to other latent driving architectures that expose a trainable scene representation. We additionally study an optional extension that grounds LAW’s predicted future latent state.

On nuScenes, structured scene-graph supervision reduces average trajectory L2 error from 0.661 to 0.622 and average collision rate from 0.456 to 0.217 relative to our controlled LAW retraining. Compared with an unstructured caption-style semantic target, it further reduces L2 error by 7.2% and collision rate by 30.4%. Our work introduces training-only relational grounding for latent driving world models without modifying inference, shows the benefit of structured relational supervision over generic textual semantics, and analyzes how

<sup>1</sup> Institute for Intelligent Systems, Esslingen University of Applied Sciences, Germany.

<sup>2</sup> Department of Computer Science, University of Freiburg, Germany.

grounding affects observed and predicted future latent states.

## II. RELATIONALLY GROUNDED LATENT WORLD MODEL

We investigate whether structured relational scene information can serve as privileged semantic supervision for the latent state of a driving world model. Building on LAW [12], we introduce an auxiliary objective to align visual scene representations with the semantic embeddings of traffic scene graphs. As illustrated in Fig. 2, we construct scene graphs from the 3D annotations for each training frame and use them only for semantic supervision. We remove all additional components at inference. We further investigate an optional temporal extension that also aligns the world model’s predicted future representation with the scene graph at time  $t + 1$ .

### A. Latent World Model

Given multi-view images  $\mathbf{I}_t$  at time  $t$ , LAW [12] uses a vision encoder  $E_{\text{vis}}$  to extract a set of visual latents,  $\mathbf{V}_t = E_{\text{vis}}(\mathbf{I}_t) = \{\mathbf{v}_t^1, \dots, \mathbf{v}_t^L\}$ . A waypoint decoder  $D_{\text{wp}}$  then predicts the future ego trajectory  $\mathbf{W}_t = D_{\text{wp}}(\mathbf{V}_t)$  from these latents. To condition the world model on the planned ego motion, the waypoint sequence is flattened into a vector using  $\text{vec}(\cdot)$  and concatenated with each visual latent, yielding the action-aware features

$$\mathbf{a}_t^i = \text{MLP}([\mathbf{v}_t^i, \text{vec}(\mathbf{W}_t)]), \quad (1)$$

where  $\text{vec}(\mathbf{W}_t)$  denotes the vectorized waypoint sequence obtained by concatenating all predicted waypoint coordinates. The resulting latent set  $\mathbf{A}_t = \{\mathbf{a}_t^i\}_{i=1}^L$  is processed by the latent world model  $F_{\text{WM}}$  to predict the future visual representation,

$$\hat{\mathbf{V}}_{t+1} = F_{\text{WM}}(\mathbf{A}_t), \quad (2)$$

which is supervised by  $\mathbf{V}_{t+1} = E_{\text{vis}}(\mathbf{I}_{t+1})$  extracted from the corresponding future observation. We denote LAW’s original trajectory and latent-prediction objectives collectively as  $\mathcal{L}_{\text{LAW}}$ . Our approach leaves this architecture and its original supervision unchanged.

### B. Scene-Graph Semantic Supervision

For each training frame, we construct a relational traffic scene graph  $\mathcal{G}_t = (\mathcal{N}_t, \mathcal{E}_t)$  from the corresponding nuScenes 3D annotations, where  $\mathcal{N}_t$  denotes the set of graph nodes and  $\mathcal{E}_t$  the set of pairwise relations. We adopt the Actor-Only representation of GraphPilot [25], in which the ego vehicle and surrounding traffic participants, such as vehicles, cyclists, and pedestrians, serve as nodes, while edges encode relations based on their relative orientation and proximity. This provides a compact representation of the relational structure of the current traffic scene.

Following GraphPilot [25], we serialize the graph into structured subject–relation–object statements, as illustrated in Fig. 3. We denote the resulting textual representation as  $s_t = \text{Serialize}(\mathcal{G}_t)$ . To obtain a fixed semantic target, we encode  $s_t$  using the frozen Qwen3-Embedding model [27],

$$\mathbf{z}_t^{\text{SG}} = E_{\text{text}}(s_t), \quad (3)$$

where  $E_{\text{text}}$  denotes the text embedding model.

In parallel, the visual latents  $\mathbf{V}_t$  are mapped into the same embedding space using a learnable alignment head  $P$ . The alignment head first performs mean pooling over the latent vectors to obtain a scene-level representation and subsequently applies an MLP projection,

$$\mathbf{z}_t^{\text{vis}} = P(\mathbf{V}_t) = \text{MLP}(\text{MeanPool}(\mathbf{V}_t)). \quad (4)$$

We align the resulting visual and scene-graph embeddings using cosine distance,

$$\mathcal{L}_{\text{SG}} = d_{\text{cos}}(\mathbf{z}_t^{\text{vis}}, \mathbf{z}_t^{\text{SG}}) = 1 - \frac{\mathbf{z}_t^{\text{vis}} \cdot \mathbf{z}_t^{\text{SG}}}{\|\mathbf{z}_t^{\text{vis}}\|_2 \|\mathbf{z}_t^{\text{SG}}\|_2}. \quad (5)$$

This auxiliary objective encourages the visual representation to capture the high-level relational semantics encoded by the scene graph. Gradients propagate through the visual encoder and alignment head, whereas the text embedding model remains frozen.

### C. Temporal Semantic Alignment

As illustrated by the right-hand alignment branch in Fig. 2, we construct the future scene graph  $\mathcal{G}_{t+1}$  from the corresponding 3D annotations and encode its serialized representation using the same frozen text encoder,

$$\mathbf{z}_{t+1}^{\text{SG}} = E_{\text{text}}(\text{Serialize}(\mathcal{G}_{t+1})). \quad (6)$$

In parallel, the predicted future latents  $\hat{\mathbf{V}}_{t+1}$  are mapped into the same semantic embedding space using the alignment head  $P$  introduced above,

$$\hat{\mathbf{z}}_{t+1}^{\text{vis}} = P(\hat{\mathbf{V}}_{t+1}) = \text{MLP}(\text{MeanPool}(\hat{\mathbf{V}}_{t+1})). \quad (7)$$

The temporal semantic alignment loss is defined analogously to Eq. 5 as

$$\mathcal{L}_{\text{temp}} = d_{\text{cos}}(\hat{\mathbf{z}}_{t+1}^{\text{vis}}, \mathbf{z}_{t+1}^{\text{SG}}), \quad (8)$$

where  $d_{\text{cos}}(\cdot, \cdot)$  denotes the cosine distance defined in Eq. 5. While LAW supervises  $\hat{\mathbf{V}}_{t+1}$  using visual latents extracted from the future observation,  $\mathcal{L}_{\text{temp}}$  additionally encourages the predicted state to preserve the relational semantics of the corresponding future traffic scene.

### D. Training Objective and Inference

The complete training objective combines the original LAW objective with the current-state and optional temporal alignment losses,

$$\mathcal{L} = \mathcal{L}_{\text{LAW}} + \lambda_{\text{SG}} \mathcal{L}_{\text{SG}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}}, \quad (9)$$

where  $\lambda_{\text{SG}}$  and  $\lambda_{\text{temp}}$  control the strength of current-state and future-state semantic supervision, respectively. All scene-graph, text-embedding, and alignment components are used only during training and discarded at inference, leaving the original LAW inference pipeline unchanged.

## III. EXPERIMENTAL EVALUATION

We evaluate whether structured scene-graph supervision improves LAW and analyze the effects of relational structure, alignment strength, and temporal semantic grounding on downstream planning performance.

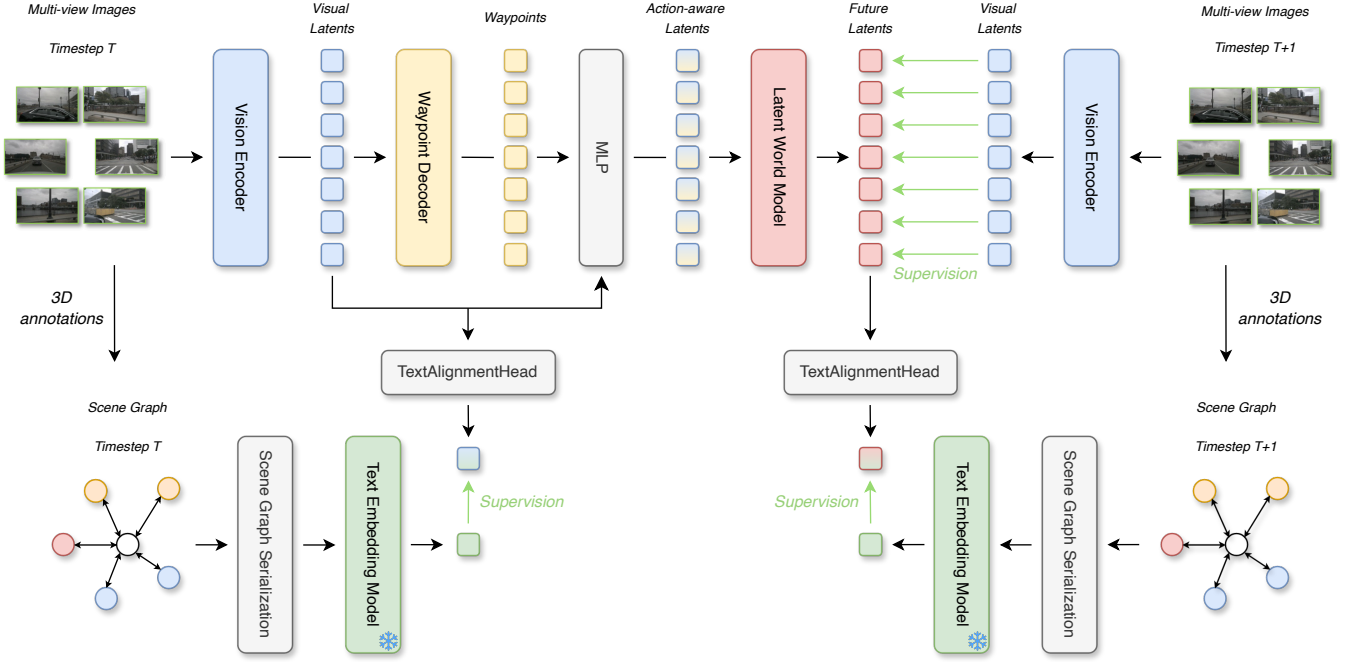


Fig. 2. Overview of our relationally grounded latent world model. The upper branch shows the original LAW [12] pipeline, which extracts visual latents from multi-view images, predicts ego waypoints, constructs action-aware latents, and predicts future visual latents. During training, we construct traffic scene graphs from the corresponding 3D annotations, serialize them, and encode them using a frozen text embedding model. We pool the visual latents at time  $t$ , project them into the semantic embedding space, and align them with the scene-graph embedding via cosine similarity. The right-hand auxiliary branch shows our optional temporal extension, which analogously aligns the predicted future latents with the semantic representation of the scene graph at  $t + 1$ . The scene-graph, text-embedding, and alignment components are discarded after training and introduce no additional inference-time cost.

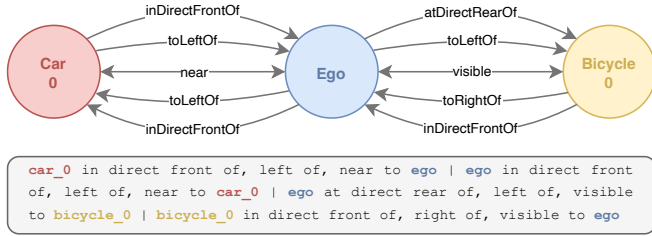


Fig. 3. Example of an Actor-Only traffic scene graph and textual serialization, reproduced from GraphPilot [25]. Nodes represent traffic participants, while edges encode spatial, proximity, and visibility relations.

### A. Experimental Setup

We evaluate our approach on the nuScenes [26] planning benchmark using the perception-free LAW architecture [12]. While LAW itself operates directly on multi-view images, our method additionally uses the available nuScenes 3D annotations during training to construct the scene-graph supervision targets described in Sec. II-B. We use these annotations only as privileged training supervision and do not require them at inference. Following LAW, we report the L2 displacement error of the predicted ego trajectory and the collision rate at prediction horizons of 1, 2, and 3 seconds, together with their averages. Lower values are better for both metrics.

For all experiments, we train LAW from scratch using the official hyperparameters and training schedule on four NVIDIA H200 GPUs. Apart from the additional semantic

supervision objective, all variants use the same LAW architecture and training setup. To ensure a controlled comparison, we retrain the LAW baseline under the same setup and report all improvements relative to this model rather than the published results or released checkpoint. Unless stated otherwise, we use the structured scene-graph serialization from Sec. II-B, a frozen Qwen3-Embedding-8B model,  $\lambda_{SG} = 0.1$ , and we set  $\lambda_{temp} = 0$  and use temporal alignment only in the corresponding ablation. We use all scene-graph and semantic-alignment components only during training and remove them at inference.

### B. Structured Relational Supervision

We first evaluate whether aligning the visual latent state with a semantic scene-graph embedding improves downstream planning. To distinguish structured relational supervision from generic textual semantic alignment, we additionally train an unstructured-text variant using the same frozen embedding model and alignment objective. Instead of explicit subject–relation–object statements, this baseline describes the scene in caption-style prose, providing the same semantic information without the explicit graph structure and relation-centric representation used by our method.

As shown in Tab. I, structured scene-graph supervision improves performance consistently across all prediction horizons. Compared with our retrained LAW baseline, the average L2 error decreases from 0.661 to 0.622, corresponding to a relative reduction of 5.9%. The improvement in collision

TABLE I

INFLUENCE OF STRUCTURED SCENE-GRAPH SUPERVISION ON NUSCENES.

| Method             | L2 [m] ↓     |              |              |              | Collision [%] ↓ |              |              |              |
|--------------------|--------------|--------------|--------------|--------------|-----------------|--------------|--------------|--------------|
|                    | 1s           | 2s           | 3s           | Avg.         | 1s              | 2s           | 3s           | Avg.         |
| LAW                | 0.327        | 0.628        | 1.029        | 0.661        | 0.156           | 0.278        | 0.934        | 0.456        |
| Unstructured Text  | 0.310        | 0.632        | 1.067        | 0.670        | 0.166           | 0.229        | 0.540        | 0.312        |
| <b>Scene Graph</b> | <b>0.287</b> | <b>0.585</b> | <b>0.994</b> | <b>0.622</b> | <b>0.098</b>    | <b>0.137</b> | <b>0.417</b> | <b>0.217</b> |

TABLE II

ABLATION OF CURRENT-STATE AND TEMPORAL SEMANTIC ALIGNMENT.

|                      | $\lambda_{SG}$ | $\lambda_{temp}$ | Avg. L2 [m] ↓ | Avg. Collision [%] ↓ |
|----------------------|----------------|------------------|---------------|----------------------|
| <i>Current-state</i> | 0.00           | 0.00             | 0.661         | 0.456                |
|                      | 0.05           | 0.00             | 0.660         | 0.283                |
|                      | <b>0.10</b>    | <b>0.00</b>      | <b>0.622</b>  | <b>0.217</b>         |
|                      | 0.25           | 0.00             | 0.696         | 0.342                |
|                      | 0.50           | 0.00             | 0.642         | 0.258                |
|                      | 1.00           | 0.00             | 0.642         | 0.312                |
| <i>Temporal</i>      | 0.00           | 0.10             | 0.628         | 0.476                |
|                      | 0.00           | 0.50             | 0.651         | 0.659                |
|                      | 0.05           | 0.05             | 0.632         | 0.257                |
|                      | 0.10           | 0.10             | 0.624         | 0.291                |
|                      | 0.25           | 0.25             | 0.606         | 0.436                |
|                      | 0.50           | 0.50             | <b>0.594</b>  | 0.387                |

rate is considerably larger, decreasing from 0.456 to 0.217, a relative reduction of 52.4%.

The comparison with unstructured text further suggests that this gain is not explained solely by introducing an auxiliary semantic target. Although unstructured semantic alignment reduces the average collision rate to 0.312, it slightly increases the average L2 error from 0.661 to 0.670. In contrast, structured scene-graph supervision improves both metrics. Relative to the unstructured representation, it reduces the average L2 error by 7.2% and the average collision rate by 30.4%. These results suggest that the structured scene-graph representation provides a more effective semantic target for the visual latent state than the unstructured textual baseline.

We hypothesize that this advantage arises from the actor-centric relational inductive bias of the scene graph. Whereas caption-style text describes scene content in a less constrained form, the scene graph explicitly emphasizes relative orientation and proximity between traffic participants, which are directly relevant to interaction-aware driving. The resulting target may therefore encourage LAW’s visual latents to preserve relational information that its original trajectory and future-latent objectives only indirectly constrain. This interpretation is also consistent with the particularly large reduction in collision rate, since collision avoidance depends strongly on representing the spatial relationships between the ego vehicle and surrounding actors.

### C. Semantic Alignment Ablations

We further analyze the influence of the semantic alignment objectives by varying the current-state loss weight  $\lambda_{SG}$  and evaluating the optional temporal alignment introduced in Sec. II-C. Tab. II presents the results.

*a) Alignment strength.*: The upper part of Tab. II shows that the effect of current-state semantic supervision is not

monotonic in  $\lambda_{SG}$ . A small weight of 0.05 already reduces the average collision rate from 0.456 to 0.283, while leaving the trajectory error almost unchanged. Increasing the weight to 0.1 yields the best overall trade-off, reducing the average L2 error to 0.622 and the collision rate to 0.217. Larger weights do not provide consistent additional gains, suggesting that scene-graph alignment is most effective as an auxiliary constraint rather than a dominant training objective.

*b) Temporal alignment.*: The lower part of Tab. II evaluates semantic supervision of the predicted future state. Future-state alignment alone improves trajectory error only at moderate weight:  $\lambda_{temp} = 0.1$  reduces average L2 from 0.661 to 0.628, but does not improve collision performance. Increasing the future-only weight to 0.5 largely removes the L2 gain and substantially worsens collision performance, indicating that overly strong supervision of the predicted latent state can be detrimental.

Joint current–future supervision exhibits a different trade-off. Increasing both alignment weights progressively improves trajectory accuracy, with  $\lambda_{SG} = \lambda_{temp} = 0.5$  achieving the lowest average L2 error of 0.594, a 10.1% reduction over the baseline. However, its collision rate of 0.387 remains substantially above the 0.217 obtained with current-state alignment alone. Overall, these results suggest that relational grounding is most effective for collision reduction when applied to the current visual state, whereas temporal grounding can further improve trajectory accuracy but requires careful weighting and does not provide the same safety benefit.

## IV. CONCLUSION

We investigated whether relational scene structure can serve as privileged semantic supervision for predictive latent world representations in autonomous driving. By aligning LAW’s visual latents with semantic embeddings of actor-centric traffic scene graphs during training, we improve both trajectory accuracy and collision performance without requiring scene graphs, 3D annotations, or additional inference computation. Our experiments further show that structured scene-graph supervision outperforms an unstructured caption-style semantic target, supporting the hypothesis that explicitly encoding relations among traffic participants provides a useful inductive bias for learning latent world-model representations. Temporal grounding of predicted future states can further improve trajectory accuracy, but it exhibits a different trade-off with collision performance.

Our current formulation obtains the semantic target by serializing the scene graph and encoding it with a frozen text model, making the target potentially sensitive to serialization choices. Future work will therefore investigate graph-native encoders, such as relational GNNs or graph transformers, that directly exploit graph topology, learn task-relevant entity and relation weighting, and provide compact relational supervision without requiring textual serialization. We further plan to study the transfer of this training-only objective to other latent driving world models beyond LAW.

## REFERENCES

- [1] Y. Hu, J. Yang, L. Chen, K. Li, C. Sima, X. Zhu, S. Chai, S. Du, T. Lin, W. Wang *et al.*, “Planning-oriented autonomous driving,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 17 853–17 862.
- [2] S. Chen, B. Jiang, H. Gao, B. Liao *et al.*, “Vadv2: End-to-end vectorized autonomous driving via probabilistic planning,” *arXiv preprint arXiv:2402.13243*, 2024.
- [3] H. Shao, Y. Hu, L. Wang, G. Song, S. L. Waslander, Y. Liu, and H. Li, “Lmdrive: Closed-loop end-to-end driving with large language models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 15 120–15 130.
- [4] C. Sima, K. Renz, K. Chitta, L. Chen, H. Zhang, C. Xie, J. Beißwenger, P. Luo, A. Geiger, and H. Li, “Drivelm: Driving with graph visual question answering,” in *European conference on computer vision*. Springer, 2024, pp. 256–274.
- [5] K. Renz, L. Chen, E. Arani, and O. Sinavski, “Simlingo: Vision-only closed-loop autonomous driving with language-action alignment,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 11 993–12 003.
- [6] X. Zhou, X. Han, F. Yang, Y. Ma, V. Tresp, and A. Knoll, “Opendrivevla: Towards end-to-end autonomous driving with large vision language action model,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 16, 2026, pp. 13 782–13 790.
- [7] F. Schmidt, K. Fedurko, M. Enzweiler, and A. Valada, “Lad-drive: Bridging language and trajectory with action-aware diffusion transformers,” *arXiv preprint arXiv:2603.02035*, 2026.
- [8] A. Hu, L. Russell, H. Yeo, Z. Murez, G. Fedoseev, A. Kendall, J. Shotton, and G. Corrado, “Gaia-1: A generative world model for autonomous driving,” *arXiv preprint arXiv:2309.17080*, 2023.
- [9] X. Wang, Z. Zhu, G. Huang, X. Chen, J. Zhu, and J. Lu, “Drivedreamer: Towards real-world-drive world models for autonomous driving,” in *European conference on computer vision*. Springer, 2024, pp. 55–72.
- [10] S. Gao, J. Yang, L. Chen, K. Chitta, Y. Qiu, A. Geiger, J. Zhang, and H. Li, “Vista: A generalizable driving world model with high fidelity and versatile controllability,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 91 560–91 596, 2024.
- [11] A. Mousakhan, S. Mittal, S. Galesso, K. Farid, and T. Brox, “Orbis: Overcoming challenges of long-horizon prediction in driving world models,” *arXiv preprint arXiv:2507.13162*, 2025.
- [12] Y. Li, L. Fan, J. He, Y. Wang, Y. Chen, Z. Zhang, and T. Tan, “Enhancing end-to-end autonomous driving with latent world model,” in *International Conference on Learning Representations*, vol. 2025, 2025, pp. 42 942–42 959.
- [13] Y. Li, Y. Wang, Y. Liu, J. He, L. Fan, and Z. Zhang, “End-to-end driving with online trajectory evaluation via bev world model,” in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 27 137–27 146.
- [14] Y. Zheng, P. Yang, Z. Xing, Q. Zhang, Y. Zheng, Y. Gao, P. Li, T. Zhang, Z. Xia, P. Jia *et al.*, “World4drive: End-to-end autonomous driving via intention-aware physical latent world model,” in *2025 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 2025, pp. 28 632–28 642.
- [15] T. Xia, Y. Li, L. Zhou, J. Yao, K. Xiong, H. Sun, B. Wang, K. Ma, G. Chen, H. Ye *et al.*, “Drivelaw: Unifying planning and video generation in a latent driving world,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 39 701–39 712.
- [16] L. Liu, Z. Song, C. Jia, H. Ye, X. Hao, L. Chen *et al.*, “Driveworld-vla: Unified latent-space world modeling with vision-language-action for autonomous driving,” *arXiv preprint arXiv:2602.06521*, 2026.
- [17] Y. Hong, X. Zhou, Y. Li, X. Zhou, L. Liu, Y. Luo, S. Xu, L. Yang, and Z. Song, “Drivefuture: Future-aware latent world models for autonomous driving,” *arXiv preprint arXiv:2605.09701*, 2026.
- [18] S. Tan, K. Chitta, Y. Chen, R. Tian, Y. You, Y. Wang, W. Luo, Y. Cao, P. Krahenbuhl, M. Pavone *et al.*, “Latent chain-of-thought world modeling for end-to-end driving,” *arXiv preprint arXiv:2512.10226*, 2025.
- [19] L. Wang, Y. Zheng, Q. Chen, S. Li, Y. Zhang, Z. Xing, Q. Zhang, X. Li, D. Qian, P. Yang *et al.*, “Latent-wam: Latent world action modeling for end-to-end autonomous driving,” *arXiv preprint arXiv:2603.24581*, 2026.
- [20] Y. Zhu, Y. Xue, H. Zhang, G. Jiang, W. Zhou, X. Yan, J. Gao, Y. Cai, B. Liu, Z. Li *et al.*, “Dlwm: Dual latent world models enable holistic gaussian-centric pre-training in autonomous driving,” *arXiv preprint arXiv:2604.00969*, 2026.
- [21] L. Zhang, C. Wu, L. Shi, J. Li, J. Liu, L. Yang, H. Zhang, M. Xu, and H. Wang, “DeepSight: Long-horizon world modeling via latent states prediction for end-to-end autonomous driving,” *arXiv preprint arXiv:2605.10564*, 2026.
- [22] Y. Luo, F. Li, S. Xu, Y. Ji, Z. Zhang, B. Wang, Y. Shen, J. Cui, L. Chen, G. Chen *et al.*, “Last-vla: Thinking in latent spatio-temporal space for vision-language-action in autonomous driving,” *arXiv preprint arXiv:2603.01928*, 2026.
- [23] L. Wild, K. Z. Luo, and M. Pavone, “Bridging structure and language: Graph-based visual reasoning for autonomous road understanding,” *arXiv preprint arXiv:2605.20942*, 2026.
- [24] Z. Song, C. Jia, L. Liu, L. Yang, S. Zhang, F. Jia, F. Zhao, P. Wu, S. Xu, C. Lv *et al.*, “Graphworld: Long-horizon planning with world models for end-to-end autonomous driving,” *arXiv preprint arXiv:2606.16274*, 2026.
- [25] F. Schmidt, M. Enzweiler, and A. Valada, “Graphpilot: Grounded scene graph conditioning for language-based autonomous driving,” *arXiv preprint arXiv:2511.11266*, 2025.
- [26] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, “nuscenes: A multimodal dataset for autonomous driving,” in *2020 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. IEEE, 2020, pp. 11 618–11 628.
- [27] Y. Zhang, M. Li, D. Long, X. Zhang, H. Lin, B. Yang, P. Xie, A. Yang, D. Liu, J. Lin *et al.*, “Qwen3 embedding: Advancing text embedding and reranking through foundation models,” *arXiv preprint arXiv:2506.05176*, 2025.